

# Charlie and the Reasoning Room

*Reason – Reinforcement Learning,  
from bandits to reasoning models*

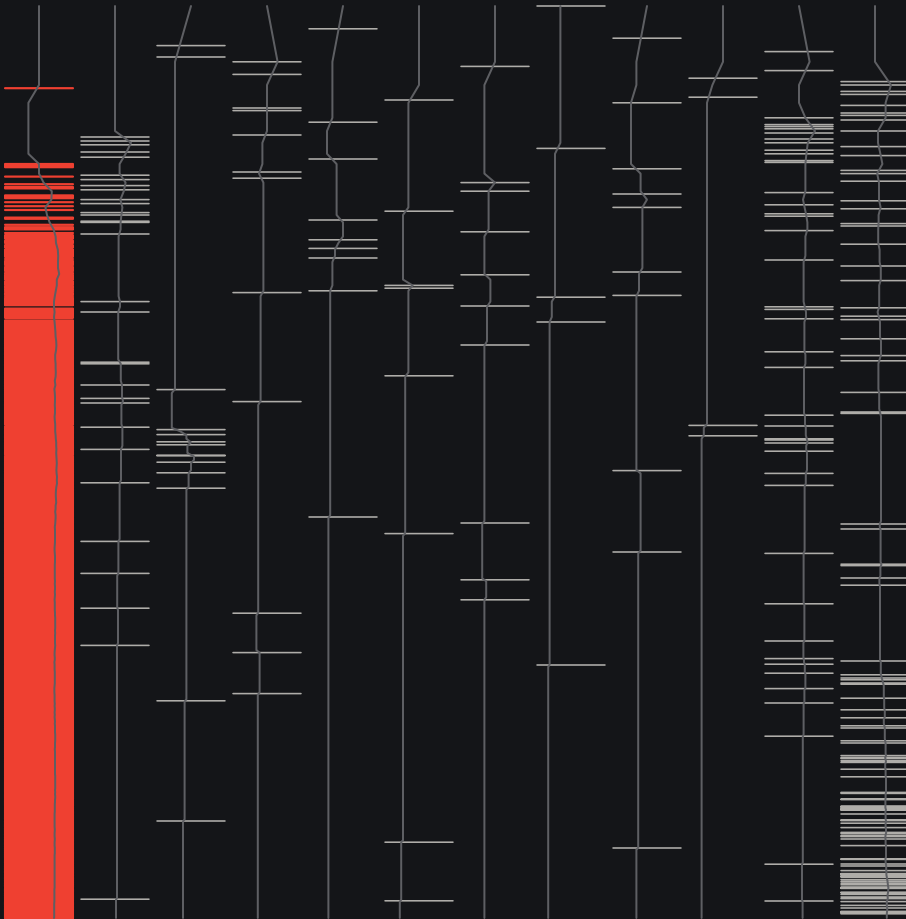


FIG. 0 Thompson sampling on a 12-armed bandit, 1,600 pulls computed in this page, time running down (on a square-root scale). Each dash is one pull; the thin line in each column is the agent's belief about that arm. The best arm ( $p = 0.72$ ) is red: it was pulled 1,345 times of the 1,600 and paid out 990.